

Question				Marking details	Marks available						
					AO1	AO2	AO3	Total	Maths	Prac	
8	(a)	(i)		it is acting as a base / it removes protons			1	1			
		(ii)		nucleophilic substitution	1			1			
		(iii)	I	M_r of —CHO fragment is 29 $\Rightarrow M_r$ of R is $72 - 29 = 43$ (1) R is C₃H₇ but branched $\Rightarrow$ compound S is H₃C H—C H₃C — C =O H (1)		1		1	2		
			II	orange / red	1			1		1	
			III	mixture melts at a lower temperature and over a greater range			1	1		1	
	(b)	(i)		use a separating / dropping funnel and run off the lower aqueous layer	1			1		1	
		(ii)		NaOH / I ₂ or KI / NaOCl (1) yellow solid (1)	2			2		2	